



AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING



Mechatronics Engineering and Automation Program Basics of mechatronics
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'The best way to predict the future is to invent it.'

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ABSTRACT

This project presents a fully functioning model of an electromechanical kid's toy enhances entertainment and engagement through interactive gameplay, suitable for the age group between 2-10 years old. The toy incorporates 10 balls, evenly split between two teams, with 5 white balls and 5 black balls. Gameplay begins upon powering the system, where balls are fed into two adjustable cannons. Players can control the cannon movement to aim at a target. Infrared sensor detects the ball's color upon successfully entering the basket, relaying information to the first microcontroller which will actively participate by announcing when a player scores, creating an interactive and engaging experience. , and simultaneously to the second microcontroller for updating the LCD screen display. Missed shots and scored balls are collected in a designated lane and lifted using an Archimedes screw mechanism. A secondary infrared sensor identifies the ball color, transmitting the data to the microcontroller, which directs a servo motor to sort the balls appropriately. The process repeats, ensuring smooth and continuous operation of the system. This setup highlights the integration of mechanical, electronic, and software components to deliver a user-friendly and interactive experience. Moreover, theoretical calculations for torque, power and projectile-based shooting mechanism were calculated to choose the suitable motor in each sub-system. The parts are designed using Inventor 2024 and printed on 3d printer or on laser cut machine. The integration of interactive gameplay features (like cannon aiming and scoring), realtime feedback via LCD and mobile applications, and modular design for easy maintenance sets it apart from conventional toys. Its use of eco-friendly materials (such as PLA filament) aligns with the growing consumer demand for sustainable products. The modular design not only simplifies maintenance but also supports scalability, making the toy viable for mass production in the future.

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Introduction

This report explores the design and development of an electromechanical interactive toy aimed at engaging children aged 2–10 years. The project integrates mechanical, electrical, and software systems to provide an immersive and educational experience. The toy features a game where players control adjustable cannons to shoot balls into a basket, fostering skills like hand-eye coordination and strategic thinking. The toy incorporates multiple subsystems: Cannon System: Adjustable cannons allow players to aim and shoot balls towards a target basket. Ball Collection and Lifting Mechanism: Missed and scored balls are collected and lifted using an Archimedes screw mechanism. Sorting Mechanism: Infrared sensors detect the color of balls and a sorting system classifies them accordingly. Scoreboard and Interface: Scores are updated in real-time on both an LCD screen and a mobile application. The design prioritizes safety, user-friendly ergonomics, energy efficiency, and environmental sustainability. By utilizing advanced tools such as Autodesk Inventor Professional 2025 for 3D modeling, Fritzing for PCB design, and Arduino for programming, the project exemplifies the application of mechatronics principles in an interactive children's toy.

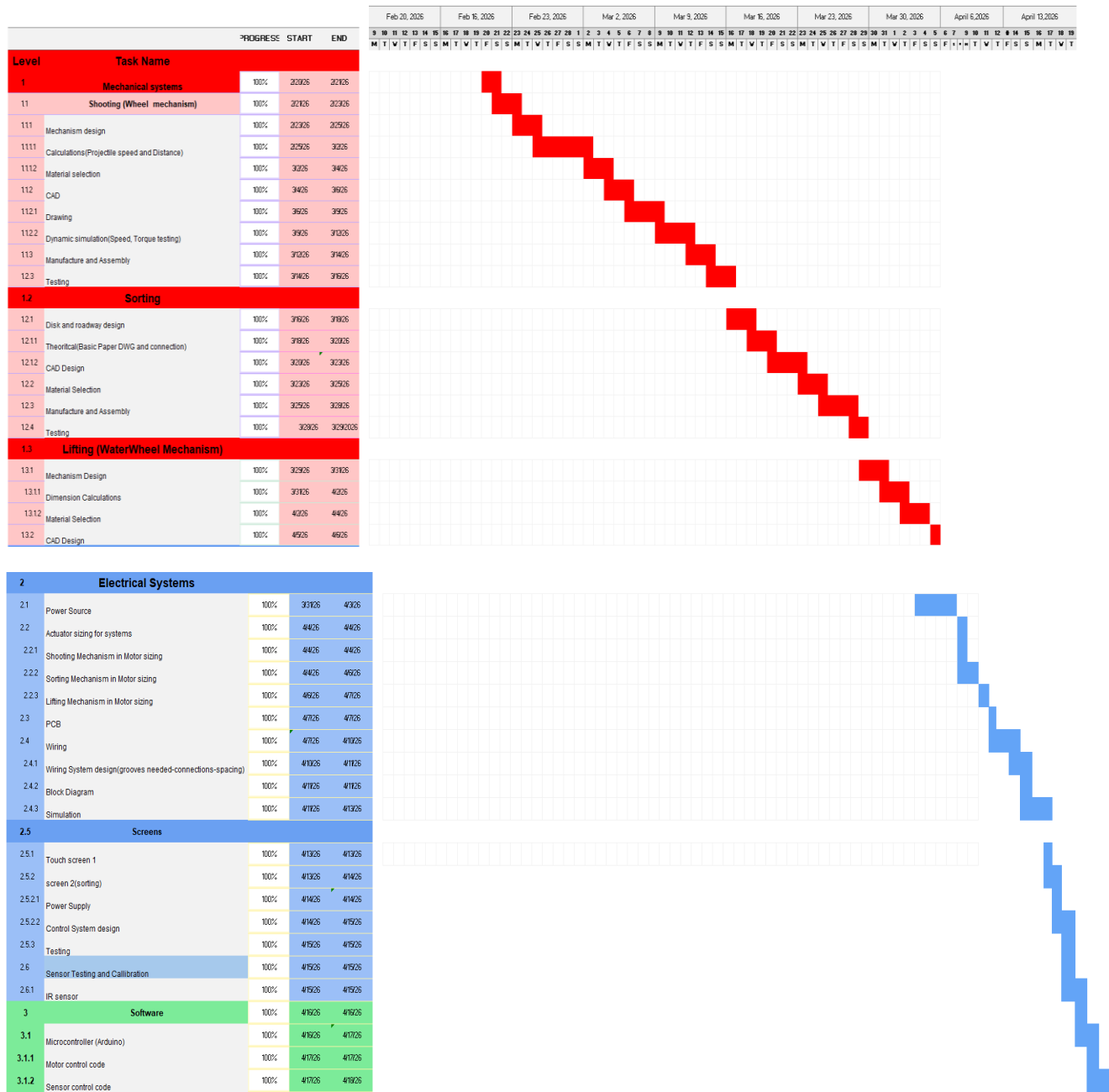
Work Breakdown Structure

WBS	
Level	Task Name
1	Mechanical systems
1.1	shooting (wheel shooter)
1.1.1	Mechanism Design
1.1.1.1	Calculations (speed- projectile calculations)
1.1.1.2	Material Selection
1.1.2	CAD
1.1.2.1	Drawing (Modelling)
1.1.2.2	Dynamic Simulation (Speed- torque testing)
1.1.3	Manufacturing and Assembly
1.1.4	Testing
1.2	Sorting
1.2.1	Disk and roadway Design
1.2.1.1	Theoretical (Basic Paper DWG's and Connections)
1.2.1.2	CAD Design
1.2.2	Material Selection
1.2.2	Manufacturing And Assembly
1.2.3	Testing
1.3	lifting (WaterWheel Mechanism)
1.3.1	Mechanism Design
1.3.1.1	Dimension Calculations
1.3.1.2	Material Selection
1.3.2	CAD Design
1.3.3	Manufacturing and Testing
2	Electrical systems
2.1	Power Source
2.2	Actuator Sizing for Systems
2.2.1	Shooting mechanism Motor Sizing
2.2.2	Sorting mechanism motor Sizing
2.2.3	Lifting Mechanism Motor Sizing
2.3	PCB Desgin
2.4	Wiring
2.4.1	Wiring system Design (groves needed- connections- spacing)
2.4.2	Block Diagram
2.4.3	Simulation on (Protus – fritzing) (testing)
2.5	Screens
2.5.1	Touch screen 1
2.5.2	Screen 2 (sorting info.)
2.5.2.1	Power Supply
2.5.2.2	Control system design
2.5.3	Testing
2.6	Sensor Testing and Calibration
2.6.1	IR sensor
3	Software
3.1	microcontroller (Arduino)
3.1.1	motor control
3.1.2	sensor control
3.2	LCD Screen programming

Gantt Chart

Fun Sphere

Project start: **Fri, 2/20/2026**



PDS (Product Design Specifications)

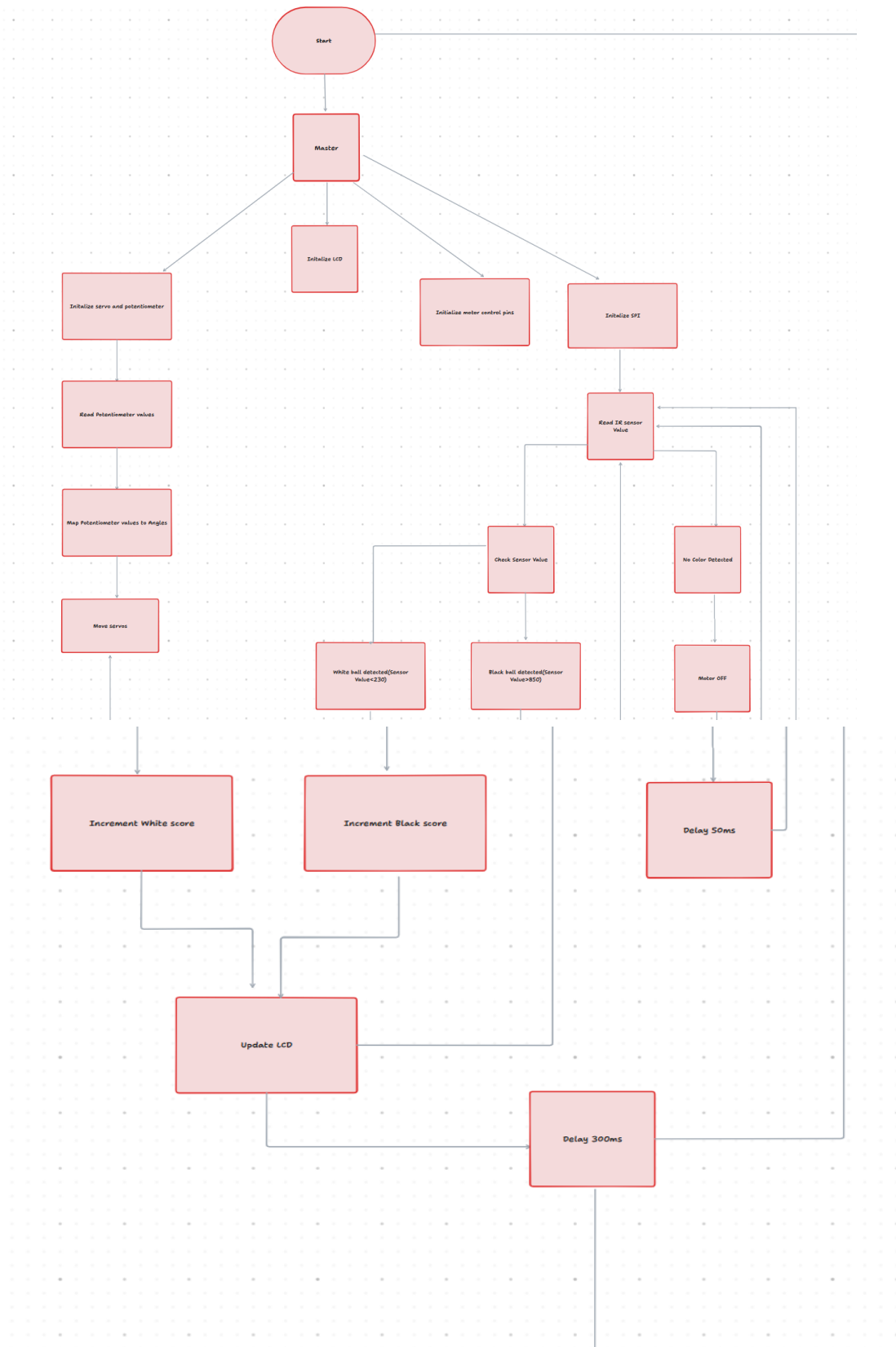
Product Design Specification (PDS): 3-Mechanism Ball Activity Centre

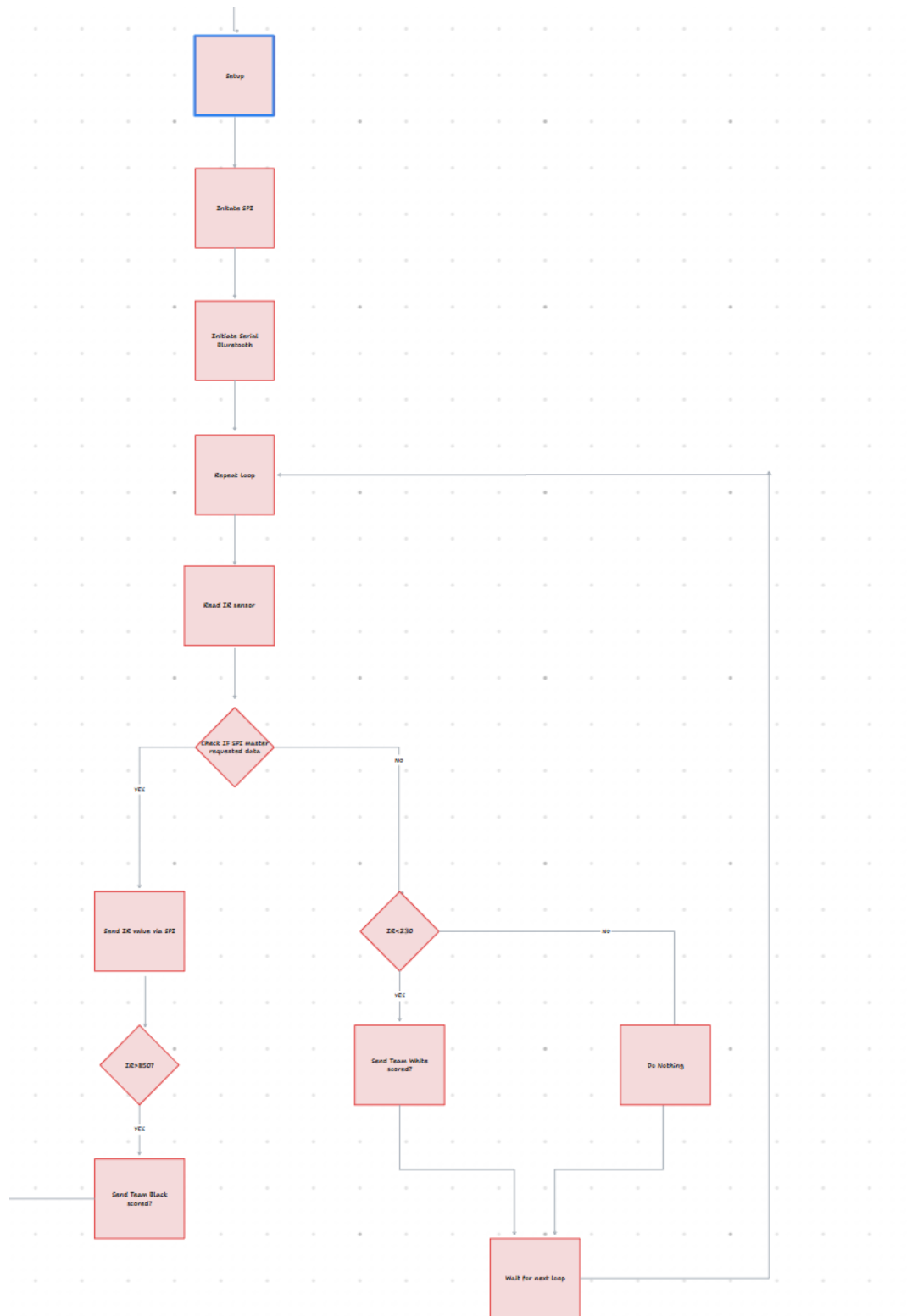
		Notes
User and Market Profile	Children aged 2–10 years (under Adult Supervision).	Design must prioritize the safety constraints of the 2-year-old (choking hazards/sharp edges) while maintaining the mechanical interest of the 10-year-old.
Operating Environment:	Primarily indoor residential or educational settings (classrooms/playrooms/Houses).	
User Interaction:	Manual interaction with LCD screens and Software UI and automated responses via sensors.	
I.Functional Requirements: The Three Mechanisms		
Lifting Mechanism:	Must transport a standard play ball (approx. 40mm diameter) from the base to a height of 15cm. Operation: motorized (DC motor).	
Sorting Mechanism:	Must differentiate balls based on a specific attribute (color). Action: Divert balls into two roadways using a stepper-actuated gate using a Sensor controlled Subsystem.	
Shooting Mechanism:	Design: dual-friction wheels that operate using DC motors.	
	Safety Constraint: Maximum launch velocity must not exceed 2 m/s to prevent injury.	

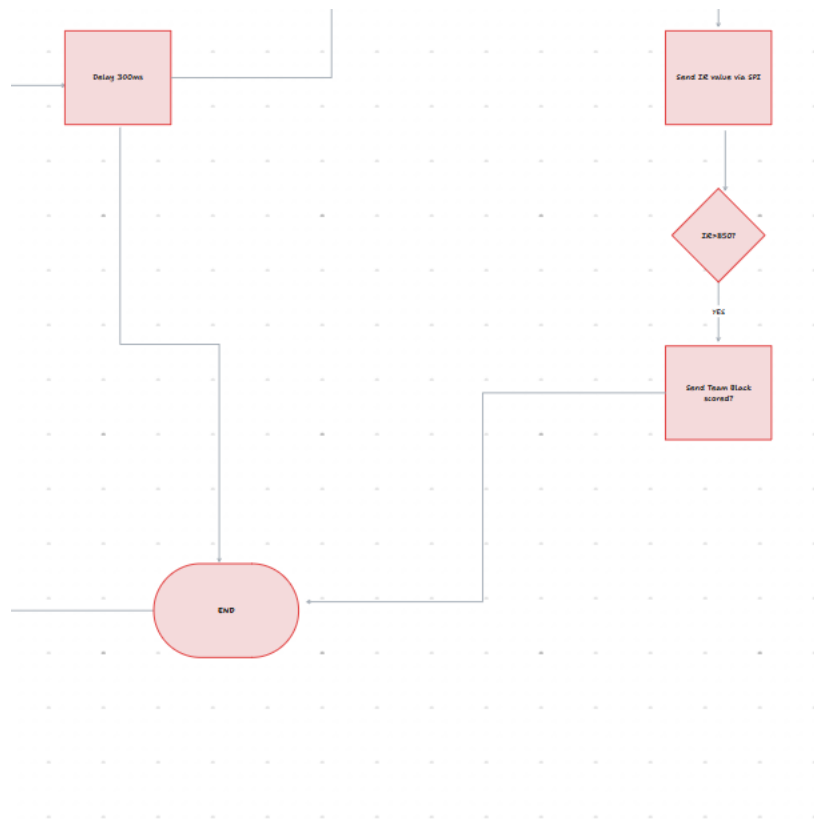
II. Safety and Regulatory Compliance		
Mechanical Safety (EN71-1 (Europe) / ES 7093 (2009) (Egypt)):		Note: The game information cards and/or packaging materials and instructions for use that are included with the game must fully and effectively draw the attention of users or their supervisors to the hazards involved in using the game and how to avoid such hazards.
Fixations:	Games, their parts, and fixing devices in the case of stationary play must have the required mechanical strength and, where necessary, stability to withstand the stresses they are subjected to during use without breaking or becoming deformed, thus potentially exposing the user to the risk of physical injury.	
Edges, protrusions, cords, cables, and connections	that can be accessed in play must be designed and assembled in such a way as to minimize the risk of physical injury from contact with them.	
Design:	must be designed and assembled to minimize the risk of physical injury that may result from the movement of different parts of the play.	
The dimensions	Maximum 1X1 meters of area of the Whole Game	
Choke Hazard:	All removable parts (balls) must be larger than the "small parts cylinder" (minimum diameter of 31.7mm). Any detachable parts must be clearly labelled for use by children under 36 months of age in a way that prevents them from being swallowed and/or inhaled.	
Edge & Corner Finish	All MDF edges must be filleted to a minimum radius of 3mm. No exposed fasteners (screws must be countersunk and plugged).	
Pinch Points	•Any moving mechanisms (gears/linkages) must be enclosed or have clearances <5mm or >12mm to prevent finger entrapment.	

3.Flammability (ES 7093 (2009) (Egypt)):		
<i>Toys must not contain flammable elements in a child's environment. Therefore, toys must be made of materials that possess the following characteristics:</i>		
<i>1.They must not ignite when directly exposed to a flame, spark, or any other potential source of fire; or</i>		
<i>2.They must not ignite easily (the flame must be extinguished once the source of the fire is removed); or</i>		
<i>3. They must burn slowly if ignited, resulting in a slow rate of flame spread; or</i>		
<i>4. They must be treated in a way that delays the combustion process, regardless of the toy's chemical composition.</i>		
4.Electrical Safety		
Voltage Limit:	Maximum operating voltage of 24V DC (Targeting 5V–9V for logic and motors).	
Battery Compartment:	: Must be secured with a screw-fastened cover to prevent child access.	
Insulation:	All soldering points and wire junctions must be heat-shrunk and housed within a non-conductive internal cavity.	
5. Materials and Environment		
Primary Material:	MDF (Medium Density Fibreboard). *Safety Grade: Must be E0 or CARB Phase 2 compliant (low formaldehyde emission) to ensure air quality safety for children. *Finish: Sealed with water-based, non-toxic, lead-free paint or clear coat (EN71-3 compliant).	
Environmental Impact:	Modular design to allow for easy disassembly and part replacement, extending the product lifecycle. Minimal use of non-recyclable adhesives; mechanical fasteners (screws/bolts) preferred for end-of-life recycling.	

Whole System Flowchart

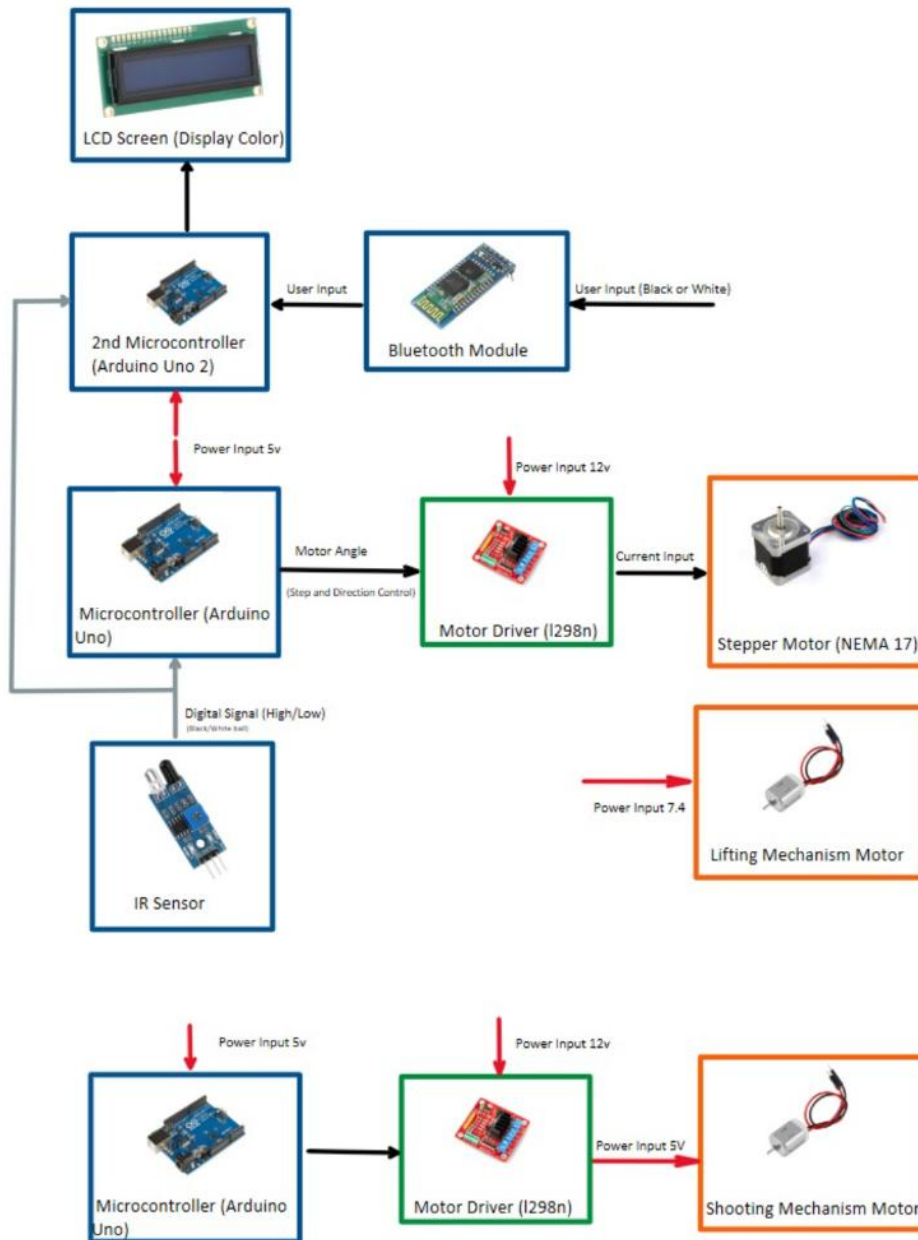






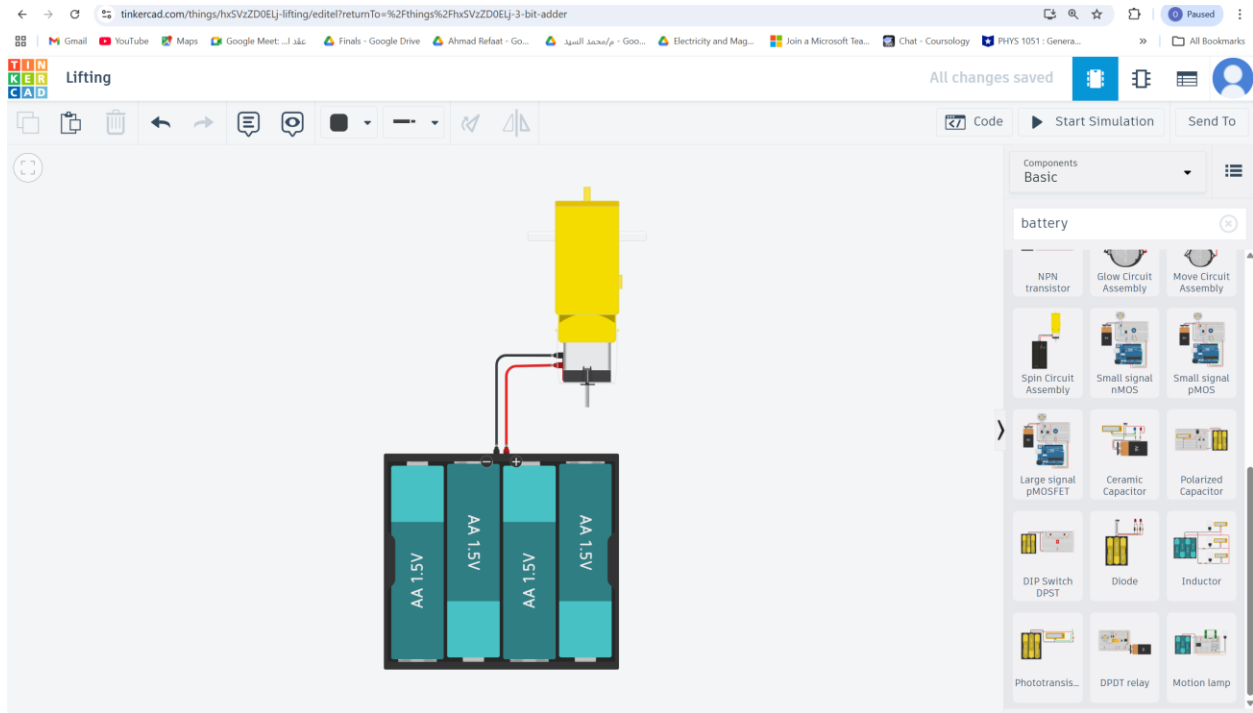
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Whole System Block Diagram

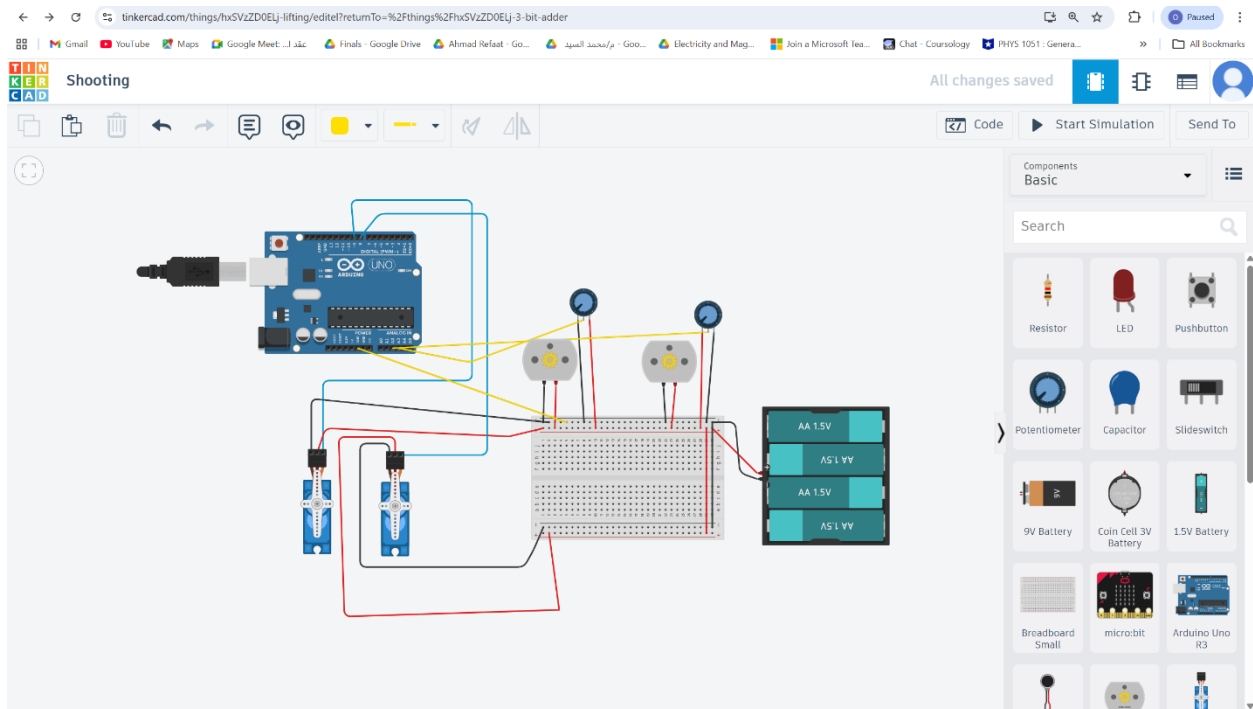


Wiring Diagram for each subsystem

Lifting



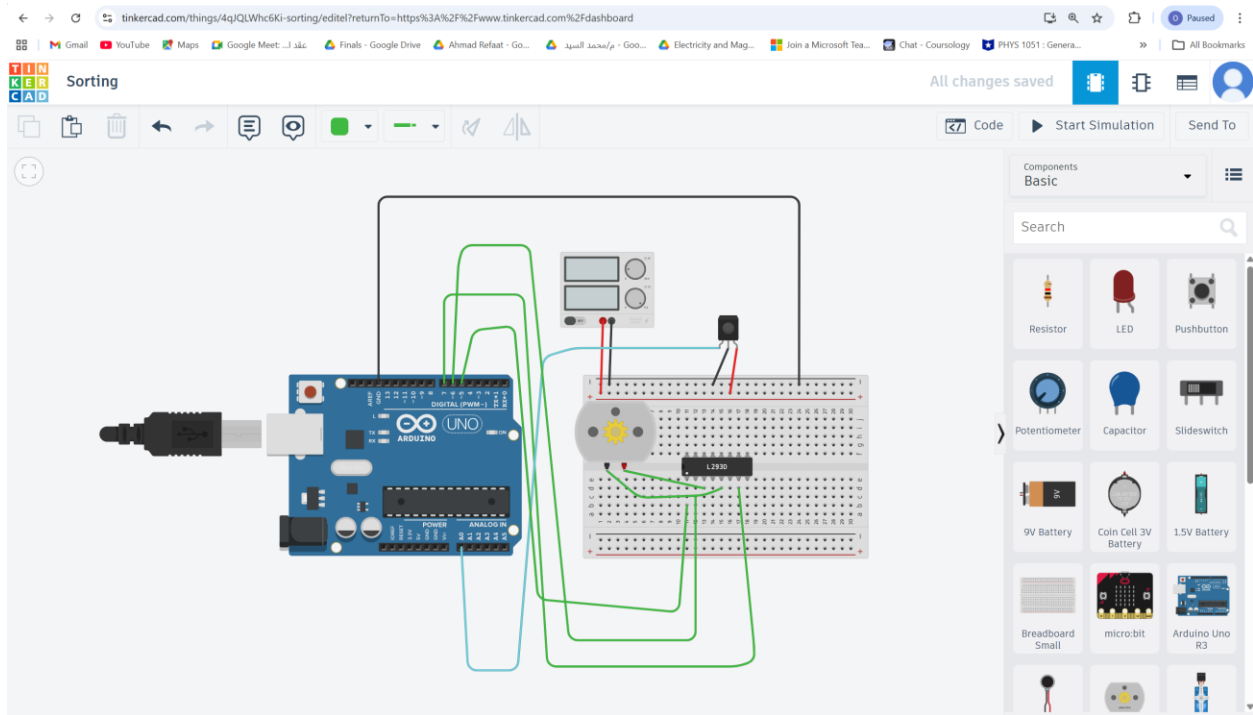
Shooting



Link

<https://www.tinkercad.com/things/hxSVzZD0ELj-shooting/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard>

Sorting

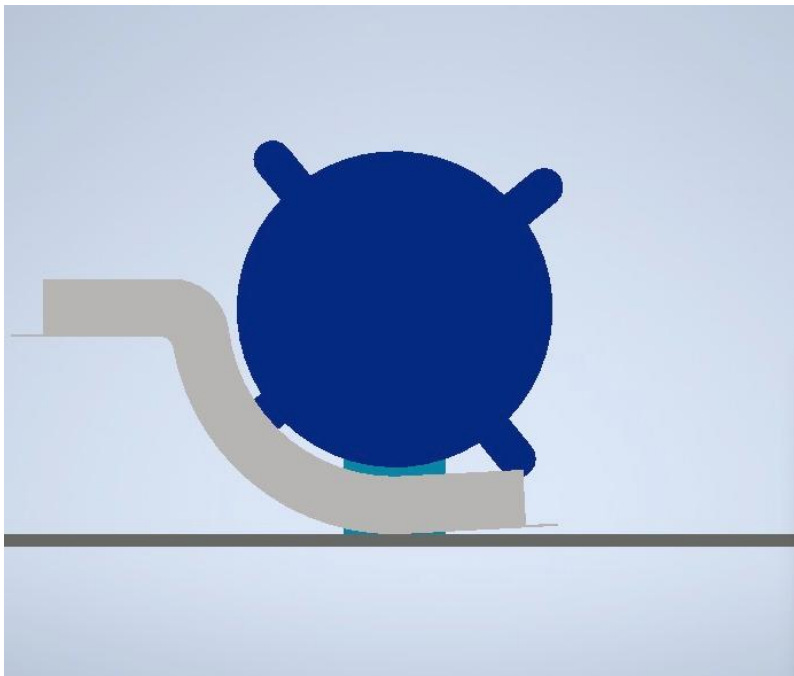
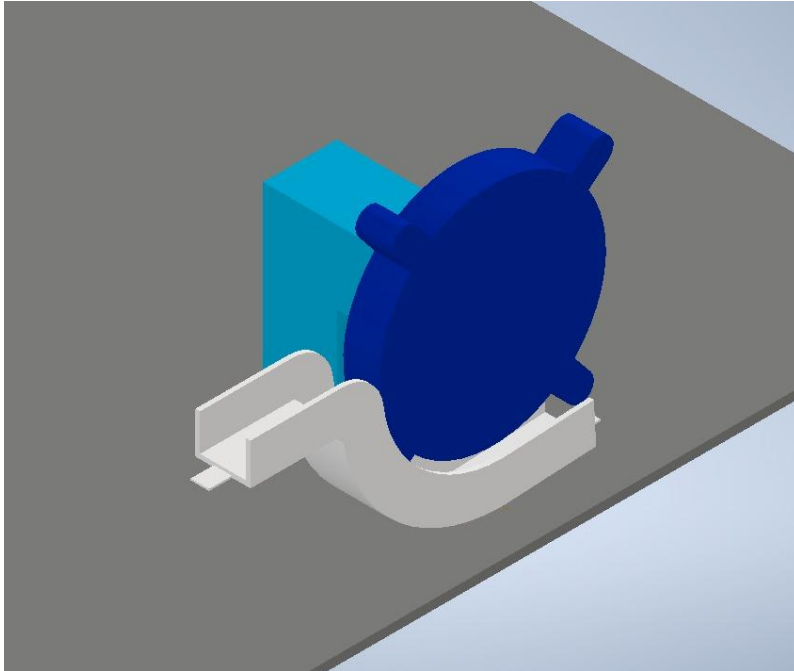


Link

<https://www.tinkercad.com/things/4qJQLWhc6Ki-sorting/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard>

Lifting

CAD Design



Actuator sizing calculations

Given Data

- Angular acceleration: $\alpha = 12.57 \text{ rad/s}^2$
- Mass moment of inertia: $J = 0.001718 \text{ kg}\cdot\text{m}^2$
- Density: $\rho = 750 \text{ kg/m}^3$
- Thickness: $t = 0.005 \text{ m}$
- Main disk diameter: $D = 0.245 \text{ m} \Rightarrow R = 0.1225 \text{ m}$
- Tabs: 4 tabs, each = rectangle $30 \times 30 \text{ mm}$ + semicircle of diameter 30 mm

1. Geometry & Volume

Disk volume:

$$V_{disk} = \pi R^2 t = \pi (0.1225^2) (0.005) \approx 2.35 \times 10^{-4} \text{ m}^3$$

Tab volume (per tab):

- Rectangle: $V_{rect} = 0.03 \cdot 0.03 \cdot 0.005 = 4.5 \times 10^{-6} \text{ m}^3$
- Semicircle: $V_{semi} = \frac{1}{2} \pi (0.015^2) (0.005) \approx 1.77 \times 10^{-6} \text{ m}^3$
- Total per tab: $V_{tab} \approx 6.27 \times 10^{-6} \text{ m}^3$

For 4 tabs:

$$V_{tabs} = 4 \cdot 6.27 \times 10^{-6} \approx 2.51 \times 10^{-5} \text{ m}^3$$

Total volume:

$$\begin{aligned} V_{total} &= V_{disk} + V_{tabs} \approx 2.35 \times 10^{-4} + 2.51 \times 10^{-5} \\ &= 2.60 \times 10^{-4} \text{ m}^3 \end{aligned}$$

2. Mass

$$m = \rho \cdot V_{total} = 750 \cdot 2.60 \times 10^{-4} \approx 0.195 \text{ kg}$$

3. Inertia Check

$$J_{disk} = \frac{1}{2} m_{disk} R^2 = \frac{1}{2} (0.176) (0.1225^2) \approx 0.00132 \text{ kg}\cdot\text{m}^2$$

4. Torque Calculation

General equation:

$$T_m = J \cdot \alpha$$

$$T_m = 0.001718 \cdot 12.57 \approx 0.0216 \text{ N}\cdot\text{m}$$

Motion Profile Analysis (Trapezoidal 1/3 – 1/3 – 1/3)

1. Timing

- Acceleration phase: $t_a = \frac{1}{3} \cdot 1 = 0.333 \text{ s}$
- Constant speed phase: $t_c = 0.333 \text{ s}$
- Deceleration phase: $t_d = 0.333 \text{ s}$
- Dwell time: $t_{dwell} = 2 \text{ s}$

- Total cycle time: $T = 3 \text{ s}$

2. Angular Displacement

linear displacement of $x = 0.4 \text{ m}$. With radius $r = 0.1225 \text{ m}$:

$$\theta_{total} = \frac{x}{r} = \frac{0.4}{0.1225} \approx 3.27 \text{ rad}$$

3. Maximum Angular Velocity

$$\theta_{total} = \omega_{max} \left(\frac{t_a}{2} + t_c + \frac{t_d}{2} \right)$$

$$\theta_{total} = \omega_{max} (0.1665 + 0.333 + 0.1665) = \omega_{max} \cdot 0.666$$

$$\omega_{max} = \frac{3.27}{0.666} \approx 4.91 \text{ rad/s}$$

4. Angular Acceleration

$$\alpha = \frac{\omega_{max}}{t_a} = \frac{4.91}{0.333} \approx 14.75 \text{ rad/s}^2$$

5. Torque per Phase

General equation:

$$T_m = J \cdot \alpha + T_{load}$$

Here:

- $J = 0.001718 \text{ kg}\cdot\text{m}^2$

A) Acceleration Phase

$$T_{acc} = J \cdot \alpha = 0.001718 \cdot 12.57 \approx 0.0216 \text{ N.m}$$

B) Constant Velocity Phase

$$T_{const} = T_{load} \approx 0$$

C) Deceleration Phase

$$T_{dec} = J \cdot (-\alpha) = -0.0216 \text{ N.m}$$

6. Peak Torque

$$T_{peak} = 0.0216 \text{ N.m}$$

7. RMS Torque

$$T_{rms} = \sqrt{\frac{1}{T} (t_a \cdot T_{acc}^2 + t_c \cdot T_{const}^2 + t_d \cdot T_{dec}^2 + t_{dwell} \cdot T_{const}^2)}$$

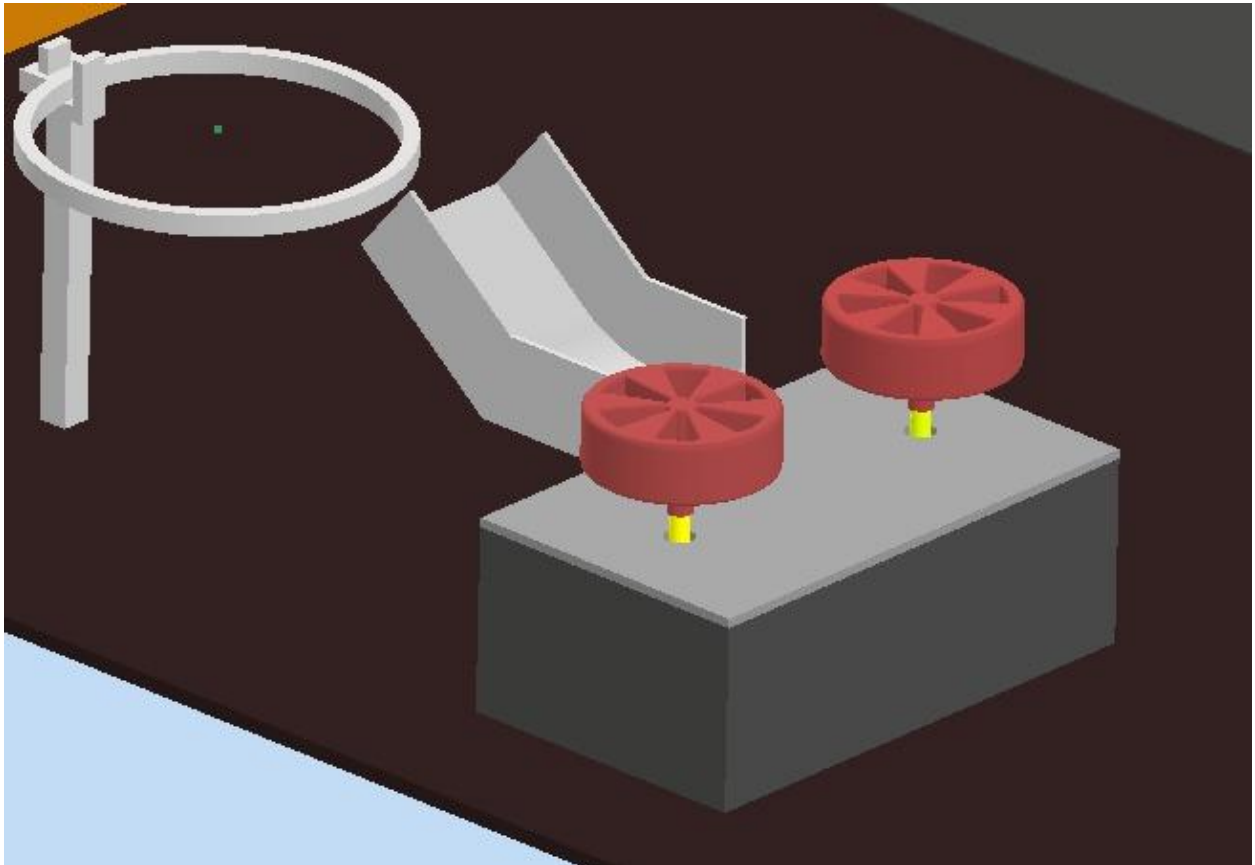
$$T_{rms} = \sqrt{\frac{1}{3} (0.333 \cdot (0.0216)^2 + 0.333 \cdot 0^2 + 0.333 \cdot (0.0216)^2 + 2 \cdot 0^2)}$$

$$T_{rms} = \sqrt{\frac{1}{3} \cdot (0.333 \cdot 0.000467 + 0.333 \cdot 0.000467)}$$

$$= \sqrt{\frac{1}{3} \cdot 0.000311} = \sqrt{0.000104} \approx 0.0102 \text{ N.m}$$

Shooting

CAD Design



Actuator sizing calculations

Given Parameters

- Mass $m = 0.015$ kg
- Radius $r = 0.06$ m
- Cycle Time = 0.48 s
- Dwell Time = 0.05 s

Inertia Calculation

Load Inertia:

$$J_{load} = m \cdot r^2 = 0.015 \cdot (0.06)^2 = 5.4 \times 10^{-5} \text{ kg}\cdot\text{m}^2$$

Motor Inertia:

$$J_{motor} = J_{load} = 5.4 \times 10^{-5} \text{ kg}\cdot\text{m}^2$$

Total Inertia (with gearbox efficiency $\eta = 0.85$):

$$\begin{aligned} J_{total} &= J_{load} + \frac{J_{motor}}{\eta} = 5.4 \times 10^{-5} + \frac{5.4 \times 10^{-5}}{0.85} \\ &= 1.175 \times 10^{-4} \text{ kg}\cdot\text{m}^2 \end{aligned}$$

Motion Profile

Rotation angle:

$$\theta_{total} = 120^\circ = \frac{120\pi}{180} = 2.094 \text{ rad}$$

Phase times (acceleration, constant, deceleration):

$$t_{acc} = t_{const} = t_{dec} = 0.14 \text{ s}$$

Angular Velocity

Equation:

$$\theta_{total} = \frac{1}{2} \omega_{max} t_{acc} + \omega_{max} t_{const} + \frac{1}{2} \omega_{max} t_{dec}$$

Substitute values:

$$2.094 = \omega_{max} \cdot (0.07 + 0.14 + 0.07) = \omega_{max} \cdot 0.28$$

$$\omega_{max} = \frac{2.094}{0.28} = 7.48 \text{ rad/s}$$

Angular Speed:

$$7.48 \text{ rad/s} \approx 71.5 \text{ RPM}$$

Angular Acceleration

$$\alpha = \frac{\omega_{max}}{t_{acc}} = \frac{7.48}{0.14} = 53.43 \text{ rad/s}^2$$

Torque Calculations

Acceleration Torque:

$$T_{acc} = \frac{J_{total} \cdot \alpha}{\eta} = \frac{1.175 \times 10^{-4} \cdot 53.43}{0.85} = 0.00737 \text{ Nm}$$

Deceleration Torque:

$$T_{dec} = -0.00737 \text{ Nm}$$

Speed Torque:

$$= 0 \text{ (no load torque considered)}$$

RMS Torque

Formula:

$$T_{rms} = \sqrt{\frac{T_{acc}^2 t_{acc} + T_{dec}^2 t_{dec}}{t_{cycle}}}$$

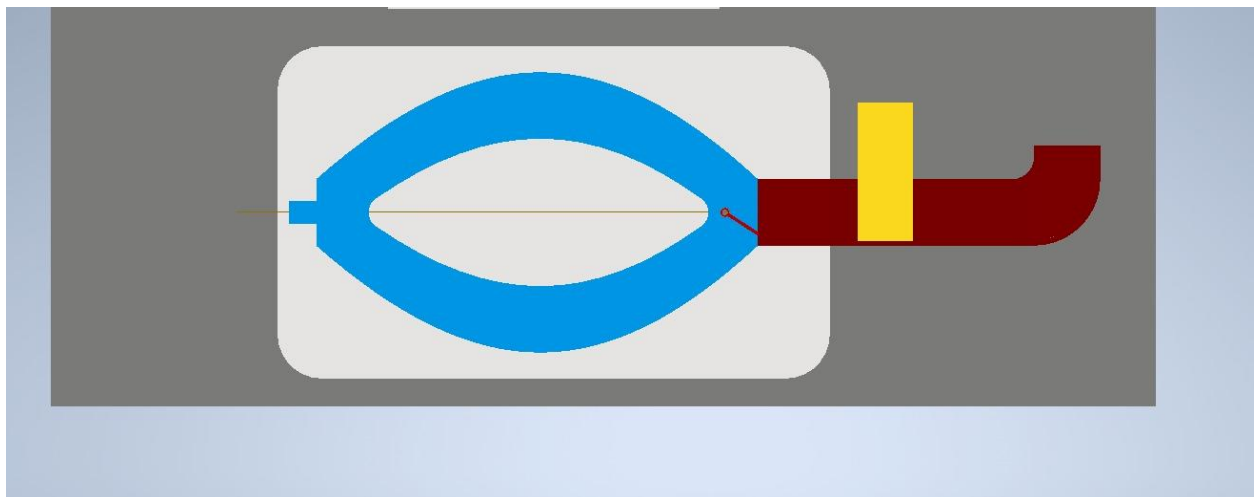
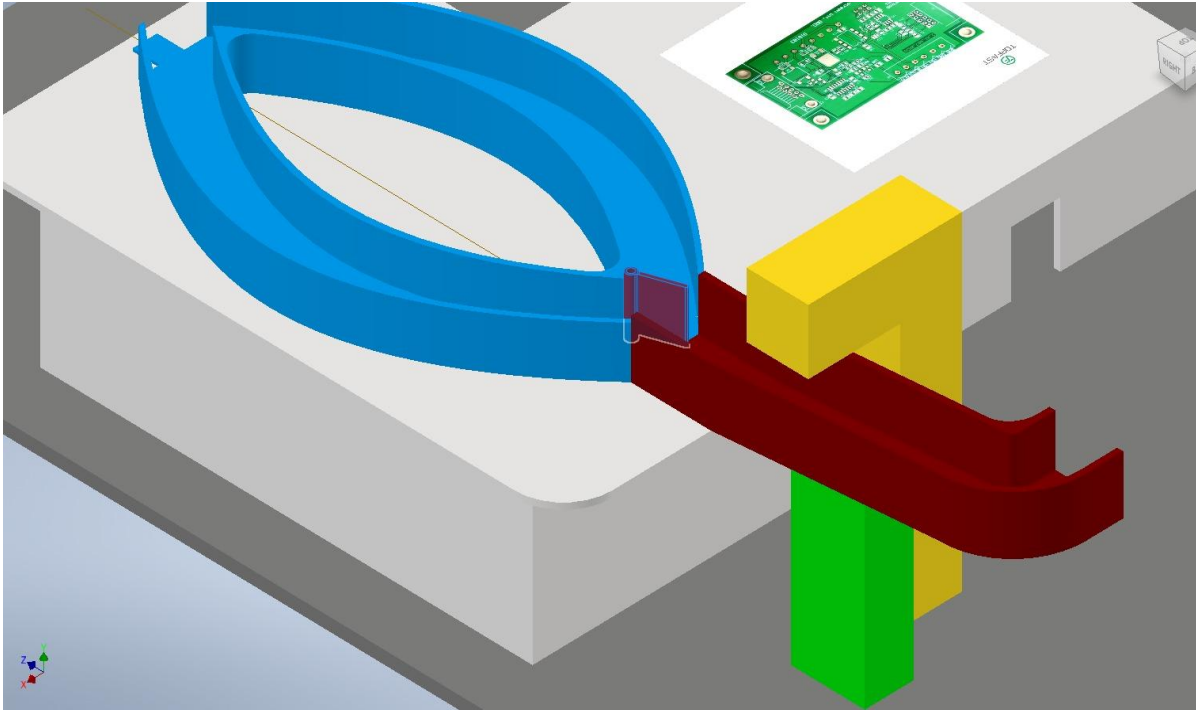
Substitute values:

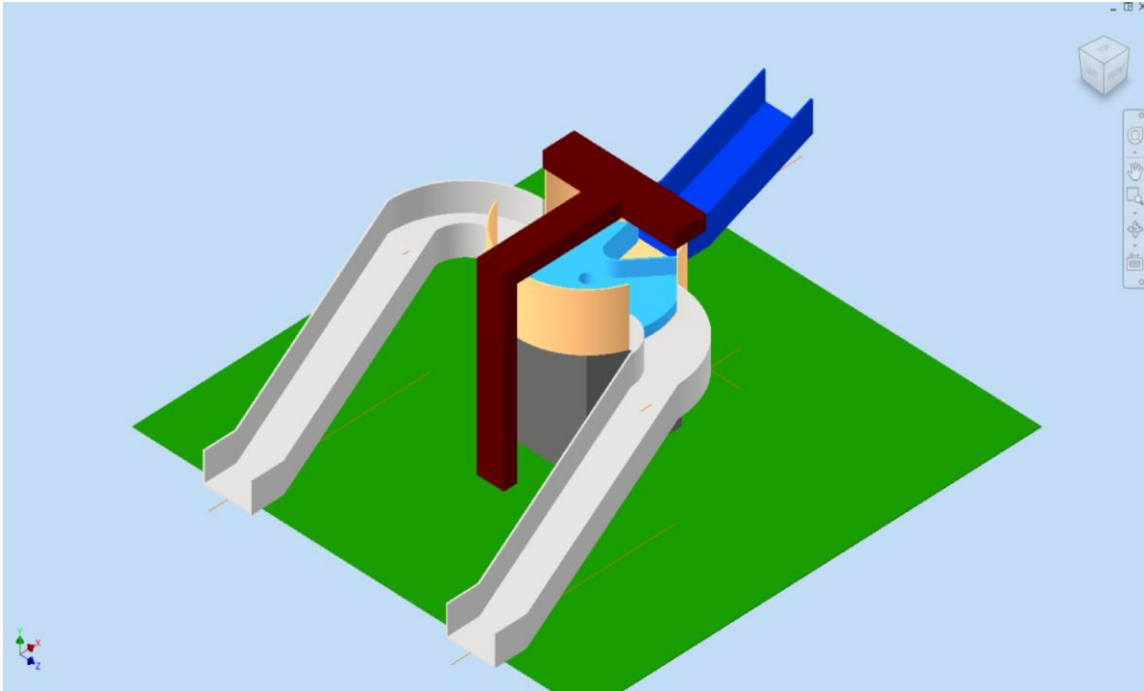
$$T_{rms} = \sqrt{\frac{(0.00737^2 \cdot 0.14 + 0.00737^2 \cdot 0.14)}{0.48}}$$

$$T_{rms} = 0.00437 \text{ Nm}$$

Sorting

CAD Design





Actuator sizing calculations

1. Servo Motor Parameters

- Motor type: NEMA 17 micro servo
- Rotor radius $r = 5mm$
- Mass $m = 0.2 \text{ kg}$

Moment of inertia of rotor:

$$J_{motor} \approx 0.5 \cdot m \cdot r^2 = 0.5 \cdot 0.2 \cdot (0.005)^2 = 2.5 \times 10^{-6} \text{ kg}\cdot\text{m}^2$$

Assume load inertia $\approx$ same order $\rightarrow$

$$J_T = J_{motor} + J_{load}$$

$$J_T = 6 \times 10^{-6} \text{ kg}\cdot\text{m}^2$$

2. Motion Profile (Rounded)

- Angular displacement: $\Delta\theta = 2\pi$ rad(one full turn)
- Total trajectory time: =6 s
- Phases:
 - Acceleration: $\alpha_{acc} = 4 \text{ rad/s}^2$
 - Const velocity= 0
 - Deceleration: $\alpha_{dec} \approx -8 \text{ rad/s}^2$

3. Torque Requirements

Peak torque:

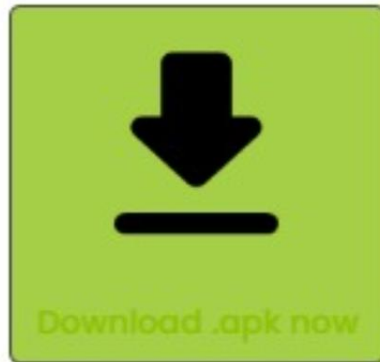
$$T_{max} \approx J_T \cdot \alpha_{max} \approx 6 \times 10^{-6} \cdot 8 \approx 5 \times 10^{-5} \text{ N.m}$$

RMS torque (averaged):

$$T_{rms} = 1 \times 10^{-7} \text{ N.m}$$

11 Mobile application

Android App for hC_05_control



Click the button to download the app, right-click on it to copy a download link, or scan the code with a barcode scanner to install.

Note: this link and barcode are only valid for 2 hours. See [the FAQ](#) for info on how to share your app with others.

Dismiss

12 Conclusion

This project successfully demonstrates the design and implementation of an interactive electromechanical toy tailored for children aged 2–10 years. The toy merges entertainment with education, allowing users to develop skills such as hand-eye coordination and problemsolving through engaging gameplay. It incorporates multiple innovative subsystems, including shooting, lifting, sorting, and scoring mechanisms, which function cohesively to create a seamless experience. In addition to its functionality, the toy adheres to ergonomic and safety standards, ensuring it is accessible and secure for children. Furthermore, its modular design enhances transportability, repairability, and sustainability by using eco-friendly materials and minimizing environmental impact.